Getting Started with Linux Fundamentals

Course | Lab Environment

Lab: Manipulating Files and Directories

Create files by using redirection, and organize files and directories.

Modify file contents, ownership, and permissions.

Outcomes

- Organize files and directories.
- Redirect command output into files.
- Use a text editor to modify files.
- Create compressed archives.
- Use sudo privileges with commands.
- Change the ownership and permissions of files.

If you did not reset your `workstation` and `server` machines at the end of the last chapter, then save any work you want to keep from earlier exercises on those machines, and reset them now.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available.

```
[student@workstation ~]$ lab start compreview-review1
```

In this exercise, you create some files on the `servera` machine. Make sure to use the exact file names from the exercise and create the files in the `/home/student` directory.

The Solution section for this lab displays command-line steps for each specification, but you can use graphical or command-line tools to perform the required tasks.

To work on this comprehensive review lab, imagine the following scenarios.

You requested a new computer and the IT Support team is ready to fulfill the request. To prepare your new workstation, the IT Support team requested some files, inventories, and back ups of the `servera` machine.

Additionally, a client asked you to organize and update some ebook files.

Specifications

- The IT Support team requested an archive called `~/user_backup.tar.gz`. This archive must contain a copy of the `/etc/passwd` file called `~/user_info.txt`, and a copy of the `/etc/shadow` file called `~/user_passwords.txt`. Remove all system accounts and leave all regular accounts in both files. If required, use sudo privileges to create the copies.

  The team also requested an inventory of the `/etc` directory. You must find all files and directories, including any items that failed the search. Save successful results in the `etc_success.txt` file, and failed results to the `etc_errors.txt` file.

  Finally, the team requested a copy of the documentation for the `mutt` email client. You can identify the documentation directory by finding a path that contains the `doc` directory. Copy all files from the resulting directory that have a `.rc` file extension to the new `~/mutt-rc` directory.

- A client requested that you organize the books in the `~/books/unsorted` directory on the `servera` machine before the books are published. Identify the category of each book by reviewing the file name. Create the necessary directories to match the following directory structure and move each book to the corresponding directory:

  ```
  books/
  ├── fiction
  │   ├── children
  │   ├── mystery
  │   └── romance
  └── nonfiction
  │   ├── dictionary
  │   ├── history
  │   └── travel
  └── unsorted
  ```

- The client also requested the following operations:

  The file name for each book includes a number; delete any file that contains a four or higher in its name.

  Change the file extension to `.pdf` for all files in the `~/books/travel` directory.

  Create the `~/books/fiction/crime` symbolic link that points to the `mystery` directory.

Copy the `children` directory with all its contents to the `nonfiction` directory.

Assign the `librarians` group as the owner of the `books` directory. Set all permissions for the owner user and the `librarians` group. Set read and execute permissions for others.

Save the total size of the `books` directory in a human-readable size unit to the `books_size.txt` file. The file must look like the following example:

```
13M
```

Show Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade compreview-review1
```

Finish

On the `workstation` machine, change to the `student` user home directory and use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish compreview-review1
```




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